#	Scenario	Learing type	Justification	Input	Output
1	Uztelecom predicts which customers will cancel their contract next month.	Supervised	Uztelecom likely has historical data of customers who canceled their contracts which can sever as labeled exmaples.	Customer data (e.g usage patterns, payment history,demographics)	Prediction of whether a customer will cancel their contract next month
2	A self-driving car learns to navigate Tashkent traffic by repeatedly trying routes and receiving penalties for unsafe moves.	Reinforcement	The car learns by trial and error and receives penalties for unsafe actions which is a classic reinforcement learning setup.	Action taken by the car (steering acceleration), sensor data.	Decisions on how to drive (steeing acceleration)
3	Uzum marketplace recommends products to users based on their browsing and purchase history.	Supervised	Uzum uses past browsing and purchase history to predict what products a user might like indicating labeled data based on user behavior.	User browsing history purchase history, history product information.	Recommended products.
4	A system groups Tashkent neighborhoods by internet usage patterns without any predefined categories.	Unsupervised	The system groups neighborhoods without predefined categories, indicating a search for inherent structure in the data.	Internet usage data for each neighborhood	Grouping of neighborhoods based on usage patterns
5	An email provider classifies incoming messages as spam or not spam.	Supervised	The email provider classifies messages based on whether they are spam or not spam indicating labeled data.	Email content, sender information	Classification as spam or not spam

6	A robot arm in a Navoi mining facility learns to sort ore by trial and error, getting a reward signal for correct picks.	Reinforcement	The robot arm learns by trial and error receiving a reward signal for correct picks fitting the reinforcement learning paradigm.	Robot arm's actions, sensor data about the ore	Sorting action
7	Dynamic pricing for Uztelecom mobile data plans — adjusting prices in real time based on demand signals and user response.	Reinforcement	The pricing adjusts based on demand signals and user response suggesting learning through interacting and feedback.	Demand signals user response to pricing changes	Price adjustment
8	A hospital in Samarkand uses patient records to predict diabetes risk.	Supervised	The hospital uses patient records to predict diabetes risking implying the existence of labeled data (patients with and without diabetes)	Patient records (ages BMI medical history)	Prediction of diabetes risk
9	An algorithm segments Uztelecom's 10 million subscribers into marketing groups based on usage behavior — no one told it how many groups to make.	Unsupervised	The algorithm segments subscribers into marketing groups to make indicating a search for inherent structure.	Subscriber usage behavior	Grouping of subscribers into marketing segments
10	A chess engine improves its play by competing against itself over millions of games.	Reinforcement	The chess engine improves by playing against itself and learning from the outcomes a clear example of reinforcement learning.	Chess game state moves made	Next move

11	Detecting SIM card fraud at Uztelecom by flagging transactions that look different from a user's normal pattern.	Unsupervised	Detecting fraud by flagging unusual transactions compared to user's normal pattern suggesting anomaly detection without labeled fraud examples.	Transaction data, user's normal transaction patterns	Flagged transaction for potential fraud
12	A news aggregator clusters articles into topics automatically.	Unsupervised	The news aggregator groups articles into topics automatically without predefined categories which is a clustering task.	Article content	Grouping of articles into topics
13	Predicting house prices in Tashkent from square footage, district, and floor number.	Supervised	Predicting house prices from features like square footage and district suggests a regression task with labeled data (historical house prices)	Square footage district floor number	Predicted house price
14	A warehouse robot learns the fastest route between shelves by exploring and receiving time-based penalties.	Reinforcement	The robot learns the fastest route by exploring and receiving time-based penalties which is characteristic of reinforcement learning.	Robot's actions time taken to reach shelves	Route between shelves
15	A model translates Uzbek text to English using a dataset of sentence pairs.	Supervised	Translating Uzbek to English using a dataset of sentence pairs is a typical supervised	Uzbek text	English translation

			learning task with labeled input-output pairs.		
16	Spotify creates a “Discover Weekly” playlist by finding users with similar listening habits and cross-recommending.	Unsupervised	Spotify finds users with similar listening habits and cross-recommends, which can be seen as clustering similar users although it could be framed as supervised.	User listening habits	’’Discover Weekly’’playlist
17	An algorithm detects unusual spikes in Uztelecom’s network traffic without labeled examples of “normal” vs “anomaly.”	Unsupervised	Detecting anomalies without labeled examples.	Network traffic data	Detected unusual spikes in network traffic.

Task 2: Reflection Questions?

1. Hard scenarios: #7, #11, #16 because they can fit multiple ML types depending on data.

2. Yes, multiple ML types can be used in one pipeline (e.g., segmentation + prediction).

3. Supervised learning is most useful due to availability of labeled telecom data